

Smart Air Filtration System Using IoT and Kalman Filter Algorithm for Indoor Air Quality and Plant Monitoring

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Abstract—Indoor air pollution is a common concern that has been causing discomfort and respiratory-related illnesses. While air filtering devices effectively solve the problem, these have high maintenance and energy consumption. As an alternative, people resort to house plants as air filters. However, it is a misconception that plants can effectively filter air pollutants in small rooms. Understanding the National Aeronautics and Space Administration (NASA) Clean Air Study, this paper proposed a sustainable system capable of filtering indoor air pollutants, specifically Carbon Monoxide (CO), Nitrogen Dioxide (NO₂), Ammonia (NH₃), and Particulate Matter 2.5 (PM 2.5). Furthermore, there is a sensing unit that aids in the automated watering of the plants. The Internet of Things (IoT) plays a vital role in efficient data processing, specifically using ThingSpeak analytics to provide real-time monitoring of the air and soil moisture levels. Considering the multiple tracking of sensing units, the implementation of the Kalman filter algorithm enables the transmission of stable data for the analysis on ThingSpeak. The design also includes comfort assessment by visualizing an Enhanced Indoor Air Quality Index on how the body perceives temperature, humidity, and indoor air pollutants. Then by performing a T-test with a probability value of 5%, the ratio of difference (t-value) between setups is determined. Comparing the design to common house plants showed a significant difference, except temperature, with all t-values greater than the critical value 1.86. In contrast, results showed the effectivity of the proposed device by having no significant difference from a commercial air filter.

Keywords— Air Filtration, IAQI, IoT, Kalman Filter, Plant Monitoring, ThingSpeak

I. INTRODUCTION

In the late 1960s, along with the increased population density and urbanization, indoor air pollution was recognized as part of the top five environmental issues with adverse effects on the human body. Commonly known as the "sick building syndrome" resulting from the accumulation of contaminants, it became one of the prevalent indoor air quality-related illnesses. A study recorded that a person spends 70 – 90% of their time indoors, making the mitigation of indoor air pollutants a topic of interest for many researchers [1]. People have considered the use of technology to maintain and monitor indoor air quality [2]. Recent advancements in technology have given rise to the various types of air filters in the market that consequently caused some concerns. A significant issue is a maintenance and cost since the filters needed to be changed and cleaned, such as high-efficiency particulate air (HEPA) filter, are expensive. The expenses also include the price for the high energy consumption while running the device. Another is the emission of ozone gas which is harmful to the human body. These are common

reasons people hesitate to purchase air purifiers or filters for their houses [3]. However, based on the study made by the National Aeronautics and Space Administration (NASA) in 1989, researchers have considered the capabilities of plants as an indoor air filter. With the application of sensor networks and smart systems, researchers suggest the potential of creating a smart air filtration system with indoor plant species [4].

Advancements in the Internet of Things (IoT) and machine learning have improved indoor air quality monitoring and plants in real-time. In a recent proposal on an IoT-based air quality monitoring device, observed contaminants include Ozone (O₃), Carbon Monoxide (CO), Particulate Matter (PM 2.5 and PM 10)—which are particles that has a diameter of smaller than 2.5 micrometers, Nitrogen Dioxide (NO₂), and Ammonia (NH₃) [5]. In designing such devices, it includes multiple sensors to measure specific gases permeating the air. Other IoT air quality monitoring systems used MQ series gas sensors, one of the most common Metal Oxide Semiconductor type gas sensors, to monitor harmful gases such as O₃, CO, and NO₂ [6, 7]. However, the data accuracy is lessened when multiple low-cost sensors accompany the system. To gather high accuracy for measured values transmitted to IoT, the study [8] suggested the implementation of the Kalman Filter algorithm, which can increase the accuracy of the sensor by 27%. These studies also involved transmitting the data from the sensors wirelessly to a remote data server. Hence, it is necessary to have a microcontroller component with a built-in Wi-Fi and Bluetooth interface. Raspberry Pi (RPi) is the microcontroller used in the previously mentioned IoT studies and satisfies the mentioned specifications. Similar to air quality monitoring, plant monitoring also involves sensors that can monitor necessary parameters. These parameters include temperature, pH, and soil moisture observed in a study similar to the topic [9]. In this proposal, integrating air quality and plant monitoring systems is requisite to a green wall structure for air purification. A green wall structure is a system that comprises plants that are arranged vertically to filtering the air of various pollutants. A study has implemented a green wall structure for improving air quality and has measured that particulate matter concentrations are constantly lower in areas nearer to the design [10]. However, the main component of plant systems that affects air quality is the plant potting mediums such as soil and carbon-based mediums. Carbon-based mediums for plants such as activated carbon aids in air filtration because of their unique properties, such as removing volatile organic compounds (VOCs), gaseous pollutants, and other odors. They are proven to increase plants' air filtration performance, growth, and prevention of molds [11]. Taking advantage of the carbon-

based potting mediums for these types of plants, they can act as an accompanying filter for the proposed system alongside the HEPA filter. The HEPA filters are common air filtration devices capable of filtering small or even smaller viruses than the COVID-19 virus [12]. A significant disadvantage of this was its maintenance cost since manufacturers do not advise the filters to be cleaned but are replaced every three to six months [13].

In order to improve air filtration and monitoring, this paper considers the increased health consciousness brought by the COVID-19 pandemic, which consequently increased the value for the air filters market. Specifically, the HEPA filter technology, having an expected compound annual growth rate of 11.0% during 2019-2023 [14]. Considering the need for air filters during the pandemic, developers should maximize the system functionality while lessening maintenance and energy consumption. Adjustments in the current indoor working conditions and health care practices can implement smart systems capable of understanding the surrounding itself [15]. The design functions on its own and could eliminate the adverse effects of the standard air filters. Considering that a standard air filter has a secondary layer other than the HEPA filter, there are still disadvantages that pose significant issues to its functionality. These issues include the low capability of these materials to filter smaller size air pollutants while maintenance should be done regularly [16]. Studies have shown materials that can effectively filter the air, heavily emphasizing the well-known NASA clean air study. However, the claim was criticized due to its effectiveness only showing significant effects in a laboratory condition. In addition to that is the minimal change in air quality, implying that plants will not efficiently filter the air independently [17]. Additives such as activated carbon must be placed in the plant's soil to maximize its air filtration capabilities [18]. Integrating these indoor plant species with activated carbon into air-cleaning technologies poses a sustainable solution to the commercial air filters, significantly improving the indoor air quality levels.

The objective of this study is to develop a smart air filtering system using IoT and implement the Kalman Filter algorithm for air quality and monitoring of plants planted on a carbon-based potting medium. Specifically, the study aims: (1) to develop a microcontroller-based system that can measure air quality and plant monitoring parameters, specifically temperature, humidity, PM2.5, CO, NH₃, NO₂ concentrations, and soil moisture; (2) to implement the Kalman Filter algorithm for the stabilization of data from sensing units; (3) to develop a Graphical User Interface (GUI) to display and record quantitative values acquired from the air and plant monitoring system; (4) To determine if there is a significant difference between the indoor air quality parameters with the smart air filtering device, without the smart air filtering device, and with a commercial air filter.

The study is a combination of implementing a monitoring and filtering system. The quantitative values obtained from the data measured by the sensor network are filtered using a machine learning algorithm, Kalman Filter. Considering that there are seven parameters monitored, it is necessary to stabilize the values from the sensing units to increase the accuracy of determining the system's actions. The design constitutes a smart plant-based system for air filtration, which is a more convenient way of filtering air than standard air filters because of the continuous monitoring of data. A smart

system would help for adjustments in an indoor working system and the efficiency of the device. It is also an implementation of a household hobby into something more beneficial. Integrating the concept of planting into an air filtering system is advantageous to resist the indoor spread of pollutants by filtering the air particles within a small area. It would also increase the performance of the plants as filters compared to [19] since the device focuses the airflow into the plant's soil where filtration occurs. Compared to other devices, this is automated to adjust fan speed at certain conditions while maintaining air quality and plants health, thus lowering energy consumption. The system is sustainable and energy-efficient compared to the commercially available air filter.

The study only covers the concept and design of a smart air filtering system with a HEPA filter that applies IoT to record the surrounding conditions, monitor the plants, and evaluate the device's effectiveness in filtering the air. The Kalman filter algorithm is only applied to filter noises in the data for the system adjustments to the surrounding conditions. The study only focuses on the filtering capabilities of the plant's carbon-based potting medium, pre-filter, and HEPA filter, excluding other types of filters. The study also excluded air conditioning or cooling of the indoor air to focus only on air filters. The soil moisture is the only variable considered in the plants' data processing, thus excluding system indications on when the plants should be changed. The researchers performed the data gathering in an indoor or room setting. Therefore, conditions in the outdoor environment do not apply.

II. METHODOLOGY

The system input involves parameters from the air and plant monitoring. The following parameters considered for air quality monitoring are temperature, humidity, gases (NO₂, NH₃, and CO), and PM2.5. For plant monitoring, the considered parameter is soil moisture. The process includes: (1) data gathering from the sensors connected to RPi; (2) data transmission from RPi to the cloud through ThingSpeak IoT Central; (3) data processing includes applying the Kalman Filter in RPi and streaming data with ThingSpeak IoT Analytics. The ThingSpeak analytics then generates a data visualization of the parameters and the enhanced indoor air quality index (EIAQI) through a dashboard constructed with ThingView. The system outputs air with a lesser concentration of air pollutants and particulate matter. It also maintains the ideal soil moisture level for plants.

A. System Process Flow

The system starts by activating the main switch to power the RPi, which starts gathering data from the indoor air and plant soil. Then the data acquired from the sensors are filtered using the Kalman Filter Algorithm. ThingSpeak performs EIAQI and soil moisture analysis based on the received filtered data. Afterward, ThingSpeak mobile dashboard displays the resulting real-time values. The user also can either choose to activate the automated system or control the system manually. Suppose the user turns on the Automated System button on the Blynk app. In that case, the automated watering and adjustment of the fan are implemented based on the EIAQI and soil moisture sensor values. If the Automated

System button is off, the user can manually control the fan speed and plant watering.

The designed smart air filtering system contains two monitoring systems: air quality and plant monitoring. The air quality monitoring system varies the fan speed depending on the measured parameters. Suppose the amount of particulate matter and gas concentrations are lower than the EIAQI level 4, the fan's speed increases at a medium level. Otherwise, the speed remains in its default value which is at a low level. The fan speed would increase its fan speed to high if EIAQI reached below 2. Suppose the value decreases, and the fan speed also lowers to the default level.

The levels at which EIAQI is categorized are the summation of Indoor Air Quality Index (IAQI) and Humidity Index (Humidex) values. These indices are used to assess the effect of indoor conditions—air pollutants for IAQI, while temperature and humidity for Humidex—in the body. The IAQI is calculated using linear interpolation with the formula.

$$I_p = I_{ll} + \left((C_p - BP_{ll}) \times \frac{I_{ul} - I_{ll}}{BP_{ul} - BP_{ll}} \right) \quad (2.1)$$

Where I_p denotes pollutants, C_p is for the rounded concentration, BP_{ul} is the breakpoint, and I_{ul} for index value.

In the case of Humidex, the value for the range is calculated using the formula.

$$h = T + \frac{5}{9} \times (6.112 \times 10^{7.5 \times \frac{T}{237.7+T}} \times \frac{RH}{100} - 10) \quad (2.2)$$

Where h equals the range, T for the air temperature in Celsius, and RH for the percent of relative humidity.

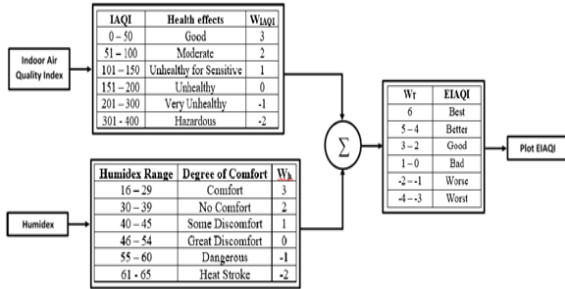


Figure 1 Enhanced Indoor Air Quality Index (EIAQI)

As shown in Fig. 1, an enhanced basis for assessing air quality was presented by combining the two indexes [20].

While for the plant monitoring system, the design has an automated watering system for the plants. If the soil moisture is less than 30%, the water pump activates to release water from the built-in reservoir of the structure. If the soil moisture exceeds 60%, the stepper is deactivated.

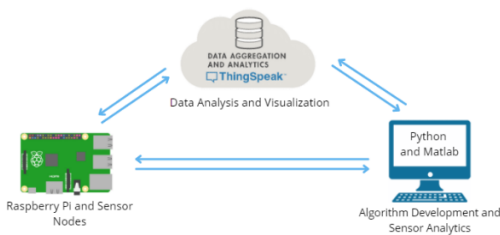


Figure 2 Data Flow of IoT System

The data flow from the sensors is shown in Fig. 2 ThingSpeak IoT Analytics is utilized for the cloud computing involved in the system. Sensor data is sent from the IoT device node—in this case, the RPi 4B—to the ThingSpeak cloud platform. Python scripts are deployed to program the conditions for air quality and plant monitoring parameters based on the data from the storage. The outputs from the Python code are sent back to the IoT device, which is significant for the air filter's automated fan speed and watering system. The streams of data are aggregated and analyzed using ThingSpeak. All data are queried, and the filtered data is displayed in the dashboard constructed using ThingSpeak visualizations and ThingView for mobile. Users can access the dashboard to view the parameters from the sensors at specific intervals of time.

B. System Design and Implementation

Objective 1: To develop a microcontroller-based system that can measure air quality and plant monitoring parameters, specifically temperature, humidity, PM2.5, CO, NH3, NO2 concentrations, and soil moisture.

1) System Architecture

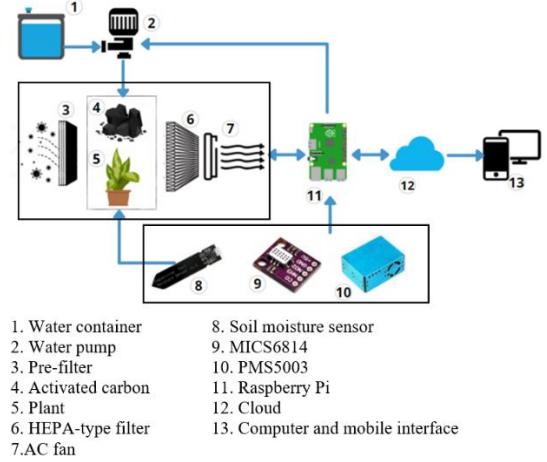


Figure 3 Architecture Diagram of Proposed System

Fig. 3 shows the architecture model of the system, taking into consideration the parts for the sensing unit, the process of air inflow, and the connection to the cloud. The soil moisture sensor, PMS5003, DHT22, and MICS6814 are the sensing units for the system. These are programmed into the RPi, which gathers the required data to be transferred to the cloud. The cloud is used to process and store the data gathered by the sensors. After data processing, the response is delivered back to the filtration system that performs specific actions. It also includes the processing of data using a Kalman filter algorithm to predict the system's behavior. Depending on the predictions, the filter can increase the power usage or add water to the potting soil. The soil sensor determines the moisture content of the potting soil, while the PM sensor and MICS6814 gather data for power usage. For the filtration system, this process includes three types of filters. Using the AC fan, the air in the atmosphere is sucked into the device, passing through the layers of filters. The first filter is a pre-filter that catches larger air pollutants. It is followed by the potting soil composed of activated carbon mixed into the soil of plants. Then the last filter is the HEPA-type filter which traps the remaining fine particles. After the

filtration process, the air is forced out into the atmosphere by the AC fan.

2) Hardware Design

The hardware involves a Raspberry Pi 4 Model B, with PMS5003, DHT22, and MICS6814 sensors. The soil moisture sensors are connected to another RPi so that the system creates an RPi cluster. The design was intentionally chosen to give way for the connection of the different modules.

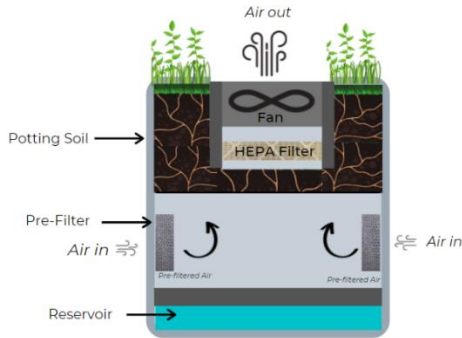


Figure 4 Prototype Internal Visualization



Figure 5 Actual Prototype Setup (a) Top View; (b) Side View

The actual prototype design is shown in Fig. 5, while the internal visualization is shown in Fig. 4. It consists of a case for the air filtration system, a container for the plant and potting soil, a water tank inside with a hose connected to the potting soil, and the filtration system itself. The dimension of the prototype is 22 by 24 inches. At the back, there are four water pathways attached to the bottom reservoir. The sensor system placed at the four sides of the plant container determines if water should be released from the reservoir through the pumps attached to the pathways. The sensor system attached at the exterior gathers data for the processing of air quality parameters.

3) Device Calibration

Sensing units require pre-calibration to allow accurate measurement of data. Values are compared to actual sensors, and the difference is calculated to determine the offset value required for the designed sensor, with an acceptable deviation of $\pm 2.0\%$.

The setup for calibrations was based on the study performed on the calibration of low-cost sensors [21]. ASTM standard was also the basis for the methods used in each calibration process. ASTM E563-11 for the temperature

sensing element and ASTM E104 - 20a for the humidity sensor. A chemical engineer was also present during the device calibration for safety reasons.

C. Preliminary Testing and Machine Learning

Objective 2: To implement the Kalman Filter algorithm for the stabilization of data from sensing units.

The Kalman Filter algorithm is implemented using the Python language in the Raspberry Pi for all sensors involved in the network. Before implementing the Raspberry Pi algorithm, the parameters are tested first in MATLAB Simulink using the Kalman Filter Block.

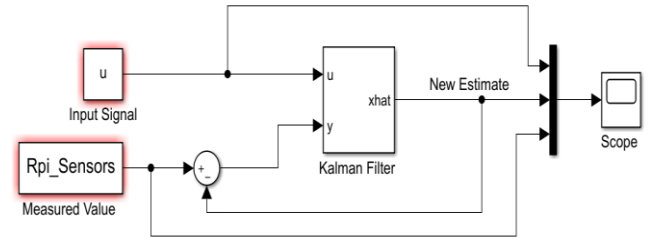


Figure 6 Simulink Model for Testing Kalman Filter Parameters

In Fig. 6, the Kalman Filter block accepts the input signal and measured values from the sensors. The block outputs an estimate based on the predicted and corrected equation sets formed on the mathematical concept of Kalman Filter. The output serves as the new estimate and feedback into the system. Data from the Raspberry Pi hardware serves as the measured values. At the same time, the input signal is a constant value from the data in the actual measuring devices. Suppose the Kalman Filter block output displays a waveform with lesser noise and closely resembles the waveform from the input signal. In that case, the input parameters (Q , R , N , and the state-space model) in the Kalman Filter block shown in Fig. 2.6 are implemented in the Raspberry Pi code. The *pykalman* library is used for applying the algorithm in Python.

D. Graphical User Interface

Objective 3: To develop a graphical user interface to display and record quantitative values acquired from the air and plant monitoring system.



Figure 7 ThingSpeak Dashboard

Fig. 7 shows a sample dashboard using ThingView applications of ThingSpeak. Data processed from the system are designed to display in the dashboard similar to the figure. It is connected to the ThingSpeak IoT cloud, which stores information from the central system. For the manual control of the system, Blynk App is integrated into the IoT device where fan speed and watering can be manually applied for cases that the user prefers to use the system in specific conditions. All information in the dashboard is updated in real-time.

E. Output Accuracy Test

The output accuracy-test involves the validation of the automated watering system. The calculated water volume based on the dimensions of the pot basin occupied by the soil is 70 mL. The amount of water dispensed was determined by the total time that the pump was active. It is calculated by first using the formula for the flow rate volume of the water. Then the volume of water to be dispensed with time is determined.

III. RESULTS AND DISCUSSION

Objective 4: To determine if there is a significant difference between the indoor air quality parameters with the smart air filtering device, without the smart air filtering device, and with a commercial air purifier

Statistical tests for the Smart air filtering system were done based on the study [22], where the air parameters were observed within 24 hours at a 3hour interval.

The air filter is placed in a small room with 5 m x 3 m x 2.5 m dimensions. To ensure the consistency of results, the researchers conducted the data gathering in a controlled environment. The room must be well-lit and have proper ventilation, so windows and doors must be closed. It must be constantly exposed to everyday household activities that produce air pollutants for each trial, as stated in [23]. Sensors were activated minutes before the stated start of recording. Before starting the data recording for the setup with air filters, air parameters were first allowed to reach a relatively similar value to the setup without an air filter.

TABLE 1. AIR QUALITY MONITORING SYSTEM PARAMETERS (WITHOUT AIR FILTERING SYSTEM)

Time	(°C)	Relative humidity (%)	PM2.5 (µg/m³)	NO2 (ppm)	NH3 (ppm)	CO (ppm)
12:00	28.0	69	165	1.1	0.91	10.3
15:00	27.3	78	100	1.3	0.84	14.4
18:00	26.1	56	95	0.9	0.70	3.3
21:00	24.7	76	24	0.7	0.42	7.6
00:00	25.6	54	77	1.2	0.20	24.2
3:00	24.3	82	34	1.4	1.03	18.6
6:00	26.0	71	125	0.9	0.15	4.2
9:00	25.8	57	48	0.9	1.38	13.2
12:00	26.9	75	109	1.3	0.64	1.1
Average	26.08	68.66	86.20	1.07	0.70	10.76

Data acquired in Table 1 shows the level of air pollutants without an air filtering system. There are also plants present in the room with similar species to the designed device. The data are from the calibrated sensors in the designed system.

TABLE 2. AIR QUALITY MONITORING SYSTEM PARAMETERS (WITH SMART AIR FILTERING SYSTEM)

Time	(°C)	Relative humidity (%)	PM2.5 (µg/m³)	NO2 (ppm)	NH3 (ppm)	CO (ppm)
12:00	27.2	69	164	1.1	0.90	10.2
15:00	25.7	70	46	0.7	0.44	3.6
18:00	25.6	49	27	0.2	0.35	0.2
21:00	25.9	57	13	0.2	0.19	3.2
00:00	24.2	61	32	0.5	0.04	10.9
3:00	25.4	65	20	0.3	0.17	8.5
6:00	25.2	54	49	0.5	0.01	1.0
9:00	26.4	46	16	0.3	0.30	5.3
12:00	27.3	64	44	0.3	0.14	0.6
Average	25.89	59.58	45.77	0.45	0.28	4.82

For the table above, testing was done in the same setup with the proposed smart air filtering system running continuously for 24 hours. Changes in levels for each air pollutants show a noticeable decrease compared to the previous setup.

TABLE 3. AIR QUALITY MONITORING SYSTEM PARAMETERS (WITH COMMERCIAL AIR FILTER)

Time	(°C)	Relative humidity (%)	PM2.5 (µg/m³)	NO2 (ppm)	NH3 (ppm)	CO (ppm)
12:00	27.0	70	165	1.1	0.90	10.1
15:00	26.8	63	45	0.6	0.39	3.3
18:00	26.1	54	25	0.1	0.26	0.1
21:00	24.9	47	9	0.2	0.22	2.8
00:00	25.8	53	30	0.4	0.00	9.9
3:00	24.1	61	21	0.4	0.15	7.3
6:00	25.0	52	46	0.3	0.07	0.7
9:00	24.7	56	12	0.3	0.24	4.6
12:00	26.9	71	38	0.3	0.12	0.4
Average	25.71	58.63	43.48	0.41	0.26	4.37

Statistical testing is applied to the means from Table 1 to Table 3. Independent samples t-test was used to determine if there is a significant difference between the different conditions. The mean differences between the air quality parameters measured in the testing of the two air filters are recorded.

Since there are nine samples for each parameter, the degree of freedom for the t-test is 16. Considering that the set p-value is 0.05, the critical value for the t-test is 1.86. The resulting t-value must be higher than 1.86 to reject the null hypothesis.

IV. CONCLUSION

In this paper, the researchers proposed a system capable of filtering air pollutants using indoor air plants, given the NASA clean air study's claim that plants can filter indoor air. The system proposed improvements to indoor plant air filters by applying IoT and Kalman filter for a smart system. It shows strengths compared to commercial air filters by providing a sustainable system with low energy consumption. In line with the fourth objective of the study, the temperature, relative humidity, PM_{2.5}, NO₂, NH₃, and CO₂ showed t-values of 0.3709, 2.0176, 1.8660, 4.9321, 2.6167, and 2.0559, respectively. Except for the temperature, results showed a significant difference between setups with and without the designed air filter. It also implies the system's ability to condition the air humidity within the environment, necessary for plant growth. For the values compared to a commercial air filter, the t-test showed no significant difference with values for temperature, relative humidity, PM_{2.5}, NO₂, NH₃, and CO₂ showing 0.3720, 0.2399, 0.1036, 0.2898, 0.1582, and 0.2365, respectively. It implies that the proposed system could perform similarly to a commercial air filter. However, the system is only limited to the areas with the same network and is relatively larger than commercial air filters. Thus, the researchers recommend further studies to improve design structure and data processing for a more comprehensive network.

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